

Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (S, FE) / S2 (PT) (S) Examination January 2024 (2019 Scheme)

Course Code: MAT206

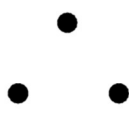
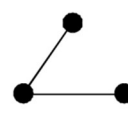
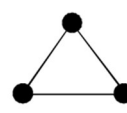
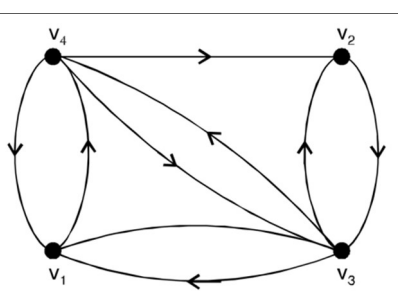
Course Name: GRAPH THEORY

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

			Marks
1		Odd number of odd vertices. Hence not possible..(3)	3
2		Drawing all 4 graphs..(3), for 3..(2), for 2 or 1 ..(1) G_1:  G_2:  G_3:  G_4: 	3
3		Drawing a tree..(3)	3
4		Definition..(1) Example ...(2)	3
5		Tree is connected. Hence a path exists..(1) If more than one path, circuits exist. Hence only one path..(2)	3
6		Binary tree has one vertex of degree 2 and n-1 vertices with degree 1 or 3..(1) Since number of odd vertices should be even..(1) n is odd..(1)	3
7		Definition..(1), Two embeddings..(1+1)	3
8		Definition..(2), Example..(1)	3
9		Chromatic number 4...(2) Justification..(1)	3
10			3

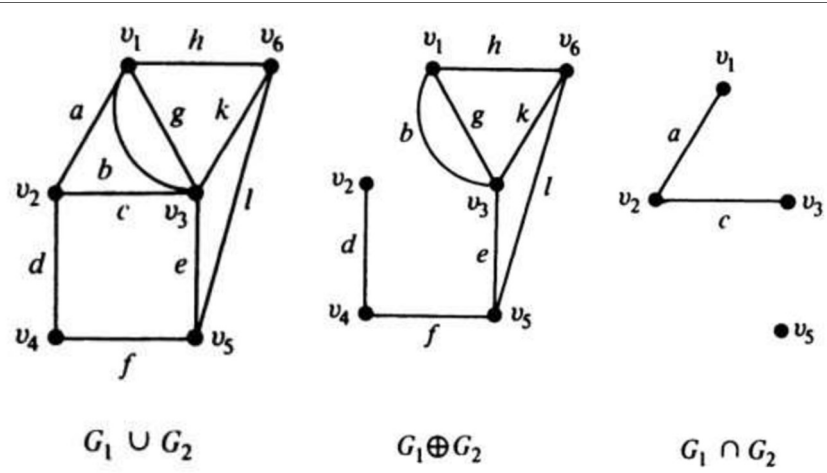
PART B

(Answer one full question from each module, each question carries 14 marks)

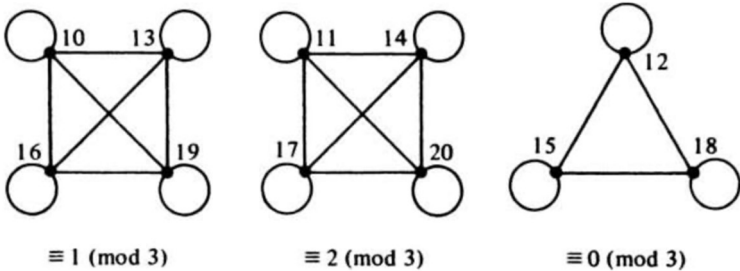
Module -1

11	a)	Definition and examples of Walk (2) Path(2), Circuit (2) Clarity ..(1)	7
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	b)	Not isomorphic..(1) Any valid argument give full credits. e.g.:- There are no circuits of length 3 in first graph.	7
12	a)	Suppose disconnected. Consider a vertex a in G and let V_1 be the set of all vertices that are joined by paths to a ..(2) Let V_2 be those vertices which are not part of V_1 . It is non empty as G is disconnected. Hence the partition..(2) Suppose partition exists..(1)..The result implies G is disconnected..(2)	7
	b)	1 mark each for K_1 to K_5 . For K_6 give 2 marks.	7
Module -2			
13	a)	If it can be decomposed into circuits, G is edge disjoint union of circuits ..(1) Hence every vertex is of even degree..(1), Hence Eulerian ..(1) Suppose G Eulerian, start with a vertex, as it is connected go to another vertex, as its degree is even move to another vertex and so on until you reach a previously traversed vertex forming a circuit(2) Remove this circuit and repeat the same process on the remaining graph. Hence the proof..(2)	7
	b)	 $G_1 \cup G_2$ $G_1 \oplus G_2$ $G_1 \cap G_2$ Ring Sum..(3), Union..(2), Intersection..(2)	7
14	a)	Complete graph has $\frac{n(n-1)}{2}$ edges...(1) Hence edge disjoint Hamiltonian circuit cannot exceed $\frac{(n-1)}{2}$... (1) Starting with an initial HC..(1), rotate the HC clockwise by $\frac{360}{n-1}, 2\frac{360}{n-1}, 3\frac{360}{n-1}, \dots, \frac{n-3}{2}\frac{360}{n-1}$ to get further $\frac{n-3}{2}$ edge disjoint HCs..(2) Hence in total $\frac{(n-1)}{2}$ edge disjoint HCs....(2)	7
	b)	Definition (1) Equivalence graph ..(6)	7

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		 $\equiv 1 \pmod{3}$ $\equiv 2 \pmod{3}$ $\equiv 0 \pmod{3}$	
Module -3			
15	a)	By induction..(1), for $n=1..(1)$, Assume up to $n=k..(1)$ Proving for $k+1..(4)$	7
	b)	Finding a minimum spanning tree with weight 24. Note edges should be chosen in increasing order of weights.	7
16	a)	To prove that there are no cycles...(1) Assume there exists a cycle. Delete an edge from the cycle...(1) This does not disconnect the graph...(1) Repeat the process in all cycles...(1) This creates a graph which is connected and circuit less. (1) A tree with less than $n - 1$ edges (1) Contradiction, hence no cycle and hence a tree..(1)	7
	b)	4 iterations and obtaining the final matrix equivalent to $\begin{bmatrix} 0 & 4 & -3 & 0 \\ -3 & 0 & -7 & -4 \\ 6 & 9 & 0 & 3 \\ 3 & 6 & -1 & 0 \end{bmatrix}$	7
Module -4			
17	a)	Suppose deletion of a cut set divides the graph into two components and a circuit lies entirely in one component. Then 0 edges in common..(2) Suppose if edges are in common, we traverse back and forth between the components as we traverse the circuit..(2) Due to closed nature of circuit, the number of edges between these components must be even..(2) Hence the proof..(1)	7
	b)	Definition (1) K_4 and dual of K_4 diagrams..(6)	7
18	a)	Edge connectivity, Vertex connectivity with examples..(2+2) separable graph...(1) Examples of separable and non separable graphs..(2)	7
	b)	Valid example give full credit	7
Module -5			
19	a)	Proof by induction..(1) True for $n=1..(1)$, Assume true for $n=k..(1)$ Consider graph G with $k+1$ vertices.	7

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		Now deleting a vertex v, we get $\kappa(G - v) \leq \Delta(G - v) + 1..(1)$ This implies $\kappa(G - v) \leq \Delta(G - v) + 1 \leq \Delta(G) + 1...(1)$ Now in G, v can have at most $\Delta(G)$ adjacent vertices ..(1) Hence $\Delta(G) + 1$ colors can be used to properly color $G..(1)$	
	b)	Suppose it contains a path of length three..(1), then the middle edge can be removed from this path without leaving the end vertices uncovered..(2) Hence not a minimal covering..(1) Conversely suppose no path of length three or more..(1) Star graphs,,(1) Hence minimal..(1)	7
20	a)	Covering..(2), Minimal Covering..(3), Covering number ..(2)	7
	b)	Circuit matrix meaning,,(1) Matrix..(6) $\begin{array}{c} e_1 \ e_2 \ e_3 \ e_4 \ e_5 \ e_6 \ e_7 \ e_8 \ e_9 \\ \begin{array}{l} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \\ z_6 \\ \underline{z_7} \end{array} \left[\begin{array}{cccccccccc} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \end{array}$	7
